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Discussion

Sui, F., and Ichchou, M. N., 2004, "Prediction of Time-Varying Vibroacoustic Energy Using a New Energy Approach" (ASME J. Vib. Acoust. 126, 184–189)

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This paper addresses the issue of transient structural dynamics in the higher-frequency range by an energetic approach. Its main purpose is to propose an alternative equation [Eq. (10) of the paper] to the usual diffusion equation [Eq. (1) of the paper] which is at the basis of the vibrational conductivity analogy of high-frequency structural dynamics. In Sec. 2.2, the authors try to justify their results (presented in Sec. 2.1) on the basis of the transport theory presented in [1] and applied to the case of randomly heterogeneous Mindlin plates in [2] or thick shells in [3]. This tentative "justification" draws several comments.

1. It is clear from transport equations (22) and (23) that W^+ and W^- depend on the wave vector $\mathbf{k}=|\mathbf{k}|\hat{\mathbf{k}}$. Then the space–time energy density $W(\mathbf{x},t)$ is obtained by integrating the half-sum of W^+ and W^- with respect to $\mathbf{k}$. The same holds for the space–time power flow density $\mathbf{I}(\mathbf{x},t)$, which is [1,2]

$$\mathbf{I}(x,t)=\frac{1}{2}c(x)\int_{\mathbb{R}^3}(W^+(\mathbf{x},\mathbf{k},t)-W^-(\mathbf{x},\mathbf{k},t))\hat{\mathbf{k}}d\mathbf{k} \tag{1}$$

if we refer to the correct theory. The phase dependence in $\mathbf{k}$ is essential because it explains the failure of the "energy constitutive law" [Eq. (5) of the paper] in multi-dimensional domains and consequently the failure of Eq. (10) to correctly represent the evolution of energy in such media. Obviously, because of this phase dependence, Eqs. (25) and (8) are not the same, contrary to what is claimed in the paper.

2. For heterogeneous media such that the group velocity $c(\mathbf{x})$ depends explicitly on the position $\mathbf{x}$, Eqs. (20) and (21)—which are in the mean time inconsistent with Eqs. (11) and (12) because $\mu\neq(\mathbb{B}\mathbf{b}_{\pm},\mathbf{b}_{\pm})$, using the notations of the paper—hold and there is no way to derive Eq. (25) or Eq. (8) from them.

3. W^+ and W^- are not equal in the undamped case and unequal in the damped case as stated in the paper, but they precisely satisfy $W^-(\mathbf{x},\mathbf{k},t)=W^+(\mathbf{x},-\mathbf{k},t)$ either in the damped or undamped cases.
Thus

$$W(\mathbf{x},t)=\int_{\mathbb{R}^3}W^+(\mathbf{x},\mathbf{k},t)d\mathbf{k},\quad \mathbf{I}(\mathbf{x},t)=c(\mathbf{x})\int_{\mathbb{R}^3}W^+(\mathbf{x},\mathbf{k},t)\hat{\mathbf{k}}d\mathbf{k} \tag{2}$$

Furthermore, the solutions of Eqs. (22) and (23) of the paper are $W^{\pm}(\mathbf{x},\mathbf{k},t)=e^{-\mu t}\tilde{W}^{\pm}(\mathbf{x},\mathbf{k},t)$ where $\tilde{W}^{\pm}$ are their solutions when $\mu=0$.

4. The damping model of Eqs. (11) and (12) is very unusual and seems to be chosen only to recover the same form of dissipation for Eq. (2) and Eq. (24) of the paper. In fact μ could simply be defined as the parameter such that the observed energy densities fit the exponential decrease predicted by the above equations. There is no need to invent a new damping model for Euler's equation (11) and the constitutive equation of barotropic fluid (12). In addition, the totally artificial introduction of frequency at this stage of the transient analysis setting $\mu=\eta\omega$ mixes time and frequency in a dubious way.

5. At last, Eq. (39) for the bending vibrations of a Euler–Bernouilli beam does not enter the framework of Eq. (13) of the paper for Friedrich's first-order hyperbolic systems.

References

[1] Ryzhik, L., Papanicolaou, G., and Keller, J. B., 1996, "Transport equations for elastic and other waves in random media," Wave Motion, **24**, pp. 327–370.
[2] Savin, E., 2002, "Transient transport equations of the energy observables in heterogeneous structures," Proceedings of the Ninth International Congress on Sound and Vibration, Orlando, FL, July 8–11. Available at http://www.onera.fr.
[3] Savin, E., 2004, "Transient transport equations for high-frequency power flow in heterogeneous cylindrical shells," Waves Random Media, **14**, pp. 303–325.

